

Practical - 4

Practical Title: JavaScript DOM Manipulation, Event Handling, and UI Interactivity

Problem Definition: Add dynamic UI components such as collapsible FAQ, modal popup, image/content slider, notification banner, hamburger menu, and light/dark theme switcher using JavaScript.

A. Theory

1. How are DOM elements selected and modified?

Answer: DOM elements are selected using methods like `getElementById()` and `querySelector()`, then modified using properties such as `textContent`, `style` and `classList`.

2. Are event listeners attached correctly?

Answer: Yes, event listeners are correctly attached using `addEventListener()` to handle user interactions like clicks, inputs, and form submissions.

3. Is localStorage used for remembering theme choice?

Answer: Yes, `localStorage.getItem('them')` and `setItem()` save and restore the darkmode preference automatically when navigating pages.

4. Does interactivity improve usability without breaking accessibility?

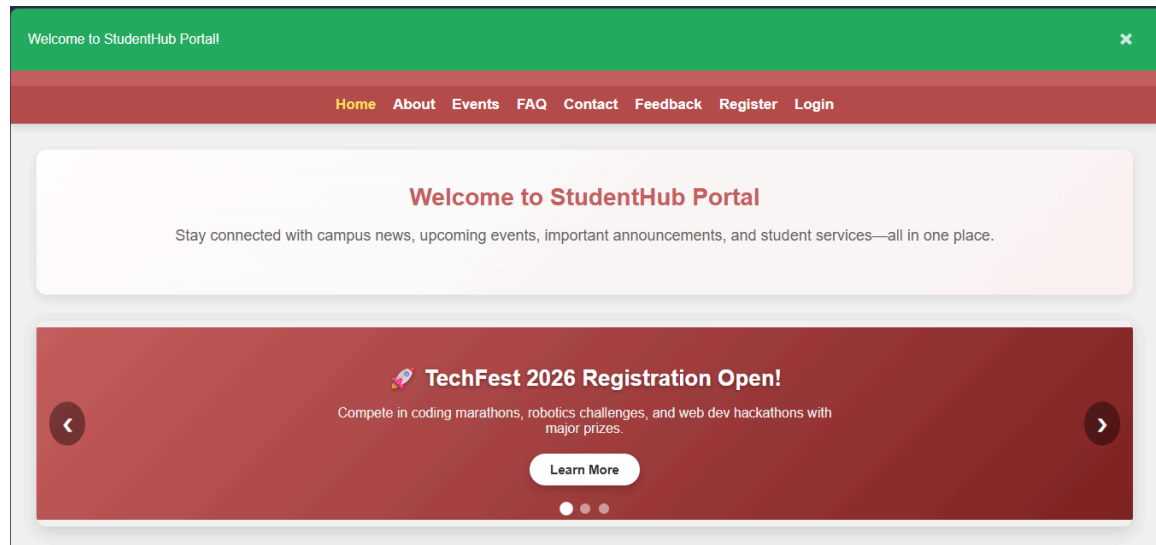
Answer: Yes, interactive components improve user experience while preserving accessibility with Escape key modal dismissal and labeled controls.

B. Implementation/ Output/Result

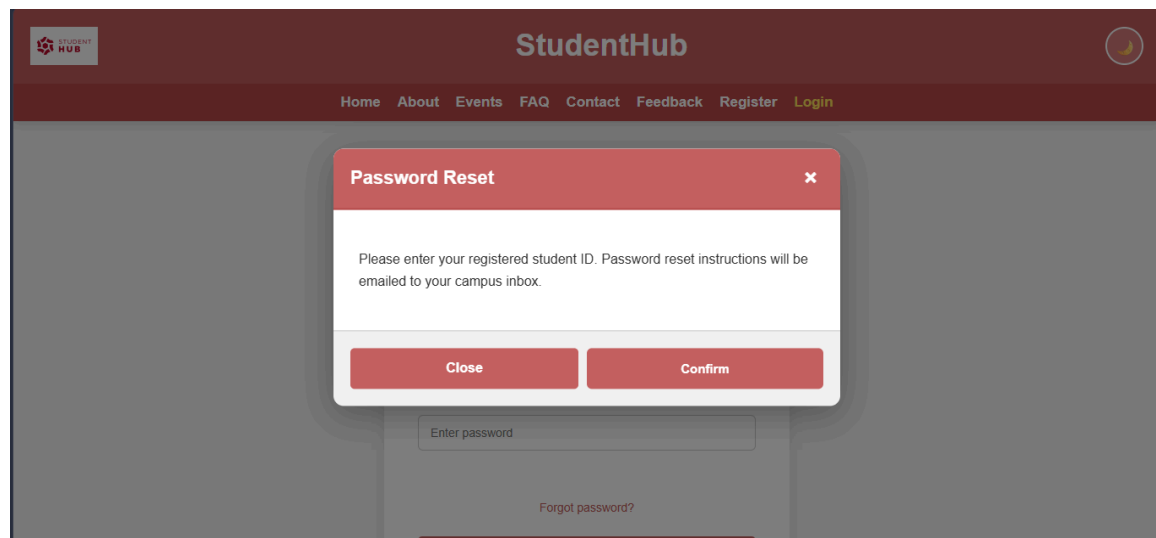
Component	JavaScript Code Snippet	Event
Theme Switcher	<code>document.body.classList.toggle('dark-theme');</code>	Persistent in LocalStorage
Accordion FAQ	<code>answer.classList.toggle('show');</code>	Auto-closes open items
Content Slider	<code>setInterval(nextSlide, 5000);</code>	Pauses on mouseenter
Modal Popup	<code>modal.classList.remove('hidden');</code>	Dismisses on Escape key

Notification Banner	<code>setTimeout(closeNotification, 5000);</code>	Triggered on form submit
Hamburger Menu	<code>navManu.classList.toggle('active');</code>	Closes on document click

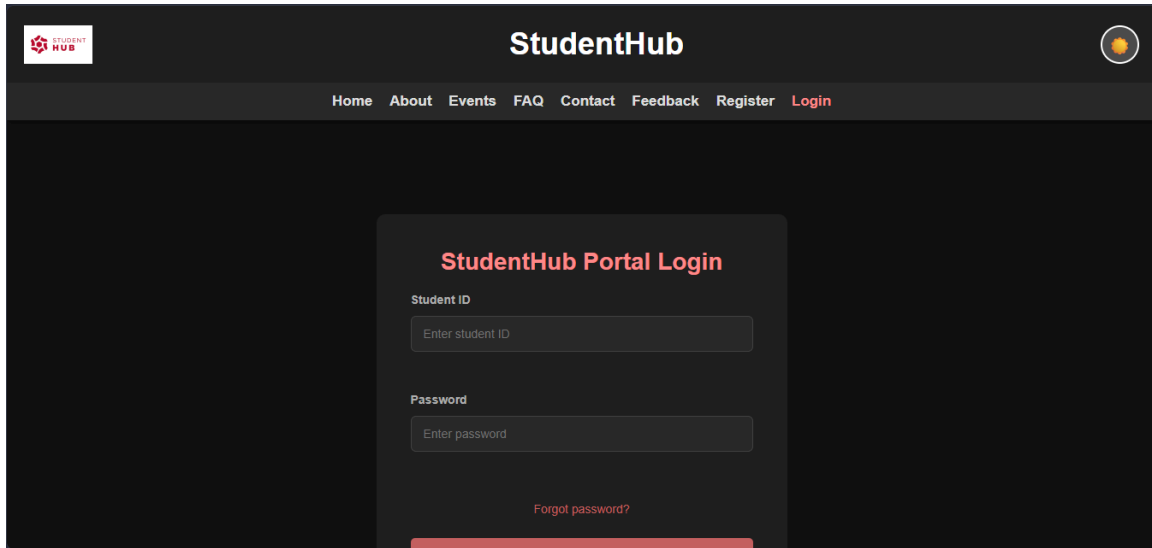
C. Evidence of Work (Screenshots)



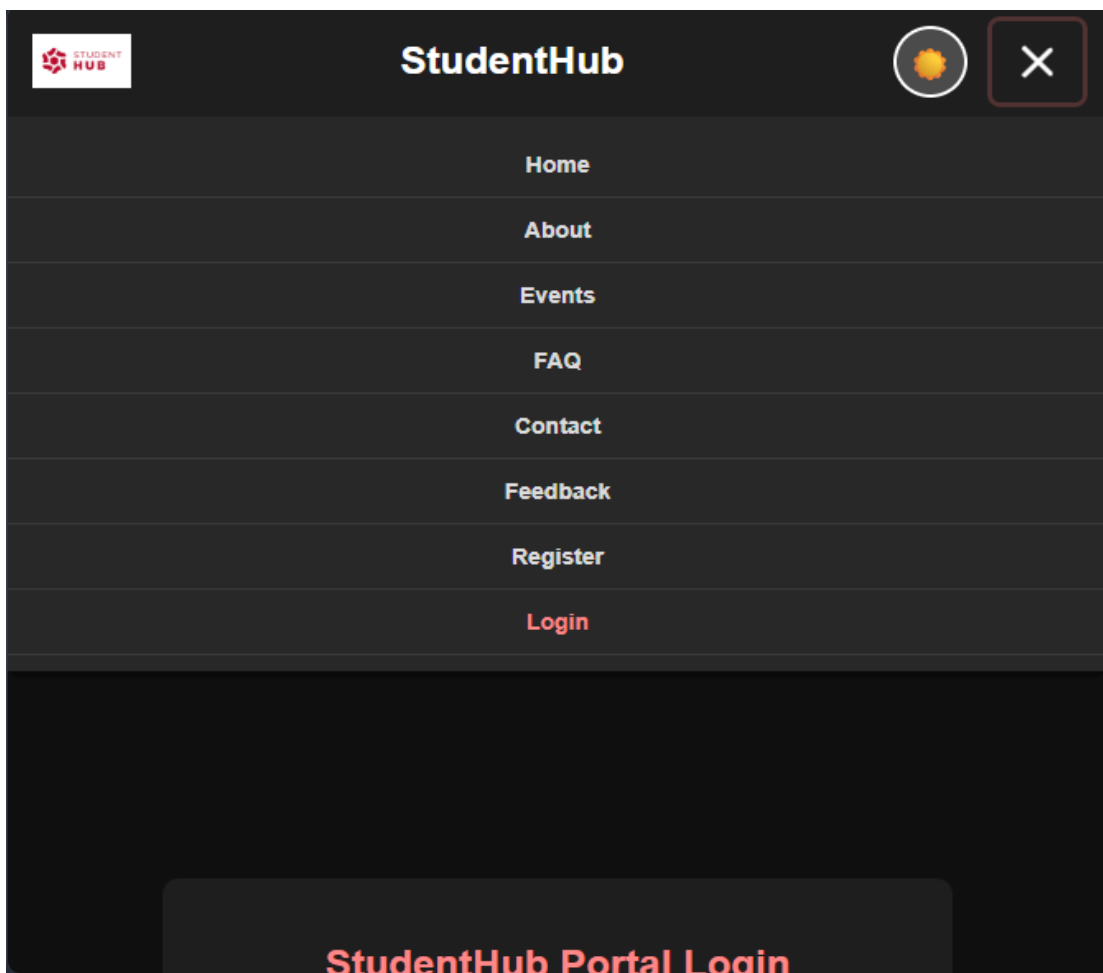
Description: Figure 4.1 Notification banner and content slider on home page



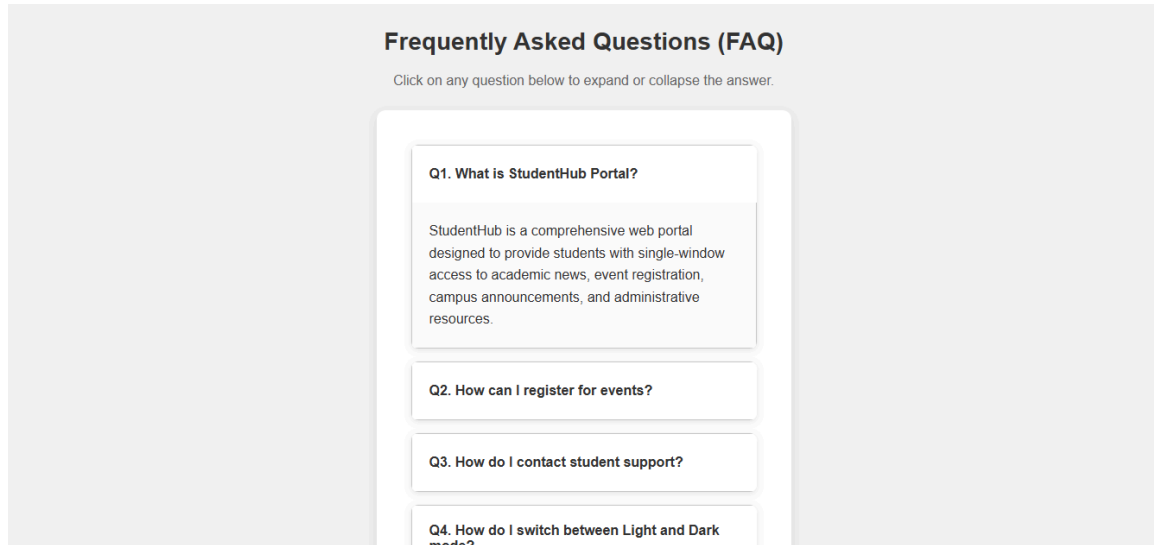
Description: Figure 4.2 Modal Popup



Description:Figure 4.3 theme switch to dark



Description:Figure 4.4 Hamburger Menu



Description: Figure 4.5 Accordion FAQ

D. GitHub Repository Link:

Link: <https://github.com/prep04-yl/StudentHub-Portal>

E. Challenges Faced:

Making sure the JavaScript code doesn't throw errors when opening pages that don't have all the form elements and getting the layout and animations to look smooth.

F. Learning Outcome:

Learned how to make web pages interactive using simple JavaScript without relying on extra libraries and understood how to save user choices like dark mode and popup helpful notifications.

Faculty Remarks: (Keep this blank for faculty)